Rare Item Arcade Escape

Game Design Notes

Version 0.4 - Working Design Document

This document records the game ideas we have agreed on so far. It is meant to be practical: something the team can read while designing and implementing the game. Anything marked as a starting value still needs playtesting.

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1. Game idea

The game is an endless arcade chase. The player controls a rare merchandise item that is trying to escape from scalpers inside an arcade. There is no final stage or win condition. A run continues until the item loses all of its protection and gets caught.

The screen is vertical (portrait). The arcade floor keeps scrolling downward, which creates the feeling that the character is moving forward through the building. The player is free to move around the playable area instead of being restricted to lanes.

The mood should be funny and chaotic, but the chase should still become stressful. The funny part comes from what is happening around the player - arcade customers, UFO catcher claws, falling coin towers, air-hockey pucks, and so on. The controls themselves should stay responsive and predictable.

2. Main gameplay loop

A normal run should feel simple at first: move through the scrolling arcade, avoid hazards, collect tickets, and try not to get pushed too far toward the scalpers. Over time the scrolling becomes faster until it reaches a cap. After that, difficulty can come from harder layouts and hazard combinations rather than making the game endlessly faster.

- New arcade sections are prepared above the visible screen and scroll downward into view.
- The player moves freely in two dimensions and chooses a route through the available space.
- Static arcade objects can block the route. They do not damage the player, but because they move with the world, they can push the player downward if the player gets stuck against them.
- Damaging hazards remove one protection layer when they hit the player.
- Tickets are optional collectibles. Some ticket paths should be safe, while others should tempt the player into taking a risk.
- Power-ups occasionally appear. The current ones are Ticket Magnet and Ticket Booster.
- Lock-on hazards such as the UFO Catcher or Key Pusher happen from time to time, but only one lock-on hazard may be active at once.
- Rare special event sections last around 10-20 seconds. During an event, the normal hazards are turned off and the event has its own hazard.
- The run ends when the bare item takes another damaging hit or is finally caught by the scalpers.

3. Screen, controls and movement

3.1 Screen and perspective

The main target is a vertical phone screen. A tall screen gives the player time to see objects entering from above while the scalpers stay below. For the art, we can still experiment with a slight fake-perspective look, but the actual movement and collision logic should remain on a simple top-down 2D plane.

3.2 Controls

On mobile, the player uses a virtual joystick. On PC/laptop, the same movement is controlled with WASD. Both versions should use the same free-movement rules so we are not designing two different games.

3.3 Movement rules

- The player can move left, right, up and down inside a fixed playable boundary.
- Movement should react quickly, with only a small amount of smoothing. We do not want slippery movement.
- The player must be able to move upward faster than the world scrolls downward. This gives the player a chance to recover after being pushed back.
- Diagonal speed should be normalized so diagonal movement is not accidentally faster than horizontal or vertical movement.
- The collision hitbox should be slightly smaller than the visible character sprite. Near-misses should feel fair.

4. Scrolling and basic difficulty

The environment, not the player, creates most of the forward-motion illusion. The character stays inside the control area while the room and its objects move downward.

4.1 Scroll speed

The game should start at a readable speed and slowly get faster. The speed must have a maximum cap. Once the cap is reached, later difficulty should come from things such as harder chunks, more demanding hazard combinations, or shorter safe openings. Exact difficulty rules after the cap can be decided later, especially when we start connecting the game design to the research paper.

4.2 Positioning risk

Staying near the top is useful because it keeps the player far away from the scalpers. However, hazards and new world objects enter from above, so the player has less reaction time there. Staying lower gives more time to see incoming objects, but it also means the player is closer to the scalper zone. This trade-off happens naturally without needing lanes.

5. Lives, damage and the scalper zone

5.1 Protection system

Instead of a normal HP bar, the item loses parts of its packaging. This gives us three usable lives while keeping the system tied to the rare-item theme:

State	What the player looks like	What happens on the next hit
1	Bag + tag + item	The bag is lost.
2	Tag + item	The tag is lost.
3	Bare item	The item is caught / Game Over.

We can add small heart icons later if testing shows that players need a clearer HUD indicator, but the bag/tag changes should still be the main visual representation.

5.2 Invincibility after a hit

After taking damage, the player gets 1.5 seconds of invincibility (I-frames). During this time another damaging hazard cannot immediately remove the next protection layer. Static blockers are still solid, so I-frames do not let the player pass through walls or machines.

A hit should not throw the character a long distance. The loss of a protection layer is already punishment enough, and a large knockback could cause an unfair second hit.

5.3 Bottom scalper zone

The bottom part of the playable area belongs to the pursuing scalpers. Reaching it does not instantly end the run if the item still has protection. Instead, a short grab animation removes one protection layer, then the player is pushed/released a little upward and receives the same 1.5-second I-frames. If the item is already bare, the grab becomes the final capture animation.

6. Obstacles and hazards

We should keep the words obstacle and hazard separate when we talk about the game. This will also make implementation discussions clearer.

Term	Meaning	Examples
Static obstacle / blocker	Does not damage the player. It blocks movement and can push the player downward with the scrolling world.	Walls, cabinets, large arcade machines
Autonomous hazard	Damages the player and follows its own movement or state pattern.	Moving person, falling coin tower
Lock-on hazard	Tracks the player first, then locks a target and attacks that fixed target.	UFO Catcher, Key Pusher
Event hazard	Only appears inside a special event section.	Air-hockey puck, clown-game balls

6.1 General warning rule

The more dangerous attacks should give the player a clear warning. For lock-on attacks, we currently plan around 3.5-5 seconds from the first warning to the actual strike. A useful starting pattern is: about 1 second of obvious warning, 2-3 seconds of tracking, then around 0.5-1 second showing that the target has locked before the attack happens. These are starting values, not final numbers.

Once a lock-on attack has committed, it should stop following the player. The challenge should be noticing the lock and moving away, not being chased by an attack that keeps correcting itself until it hits.

6.2 UFO Catcher

1. A warning appears from either the left or right side of the screen.
2. The claw tracks the player on the Y-axis for a short time.
3. The Y-position locks.
4. A clear sound/visual cue tells the player that the lock has happened.
5. The claw extends horizontally along the X-axis at the locked Y-position.
6. If it touches the player, one protection layer is lost.

6.3 Key Pusher

7. A keyhole/target effect appears around the player.
8. The target follows the player for a short time.
9. The target locks to the player’s current position.
10. The player gets a brief final warning and must leave the marked area.
11. The pusher/key smashes the locked location.
12. A hit removes one protection layer.

The UFO Catcher and Key Pusher are both lock-on hazards, so they should never be active at the same time.

6.4 Moving person

For the first version, only one moving person should be active at a time. The person moves at a normal readable speed, can cross left-to-right or right-to-left, may stop for up to around 2 seconds, and may reverse direction once. Touching the person removes one protection layer.

6.5 Coin tower

The coin tower first looks like a normal static object. Before it falls, it starts wobbling. It should then lean enough to show which side it is going to fall toward. The player should be reacting to a readable warning, not guessing left or right. Being hit by the falling tower removes one protection layer.

7. Tickets, score and power-ups

7.1 Score and distance

Competitive score should come from progress through the scrolling world, not from ticket collection. Ticket placement can vary from run to run, so using tickets for the leaderboard would make the score too dependent on luck.

The basic calculation we are using is:

distance += ScrollSpeed x DeltaTime

score = floor(distance x k)

The value k is only a display/scaling value so that the score looks nice. Since distance comes from scroll speed over time, surviving a faster part of the run naturally adds progress faster. We can show both Distance and Score on the results screen even though they come from the same progress measurement.

7.2 Tickets

- The visible ticket counter starts from 0 at the beginning of every run.
- Tickets collected during the run are added to a run total.
- At Game Over, that run total is added to the saved ticket wallet.
- The full saved wallet does not need to stay on the gameplay HUD. It can mainly appear in the menu/shop.
- Saved tickets will be used for in-game purchases such as skins, different bags/tags, and other cosmetic items. The exact shop system will be designed later.

Tickets should not be scattered as completely random single objects. We can use small patterns - straight lines, diagonal trails, curves, or risky paths through narrower openings. Some patterns should tempt the player to take a harder route, but tickets should never be placed in genuinely impossible positions.

7.3 Current power-ups

Power-up	Current idea
Ticket Magnet	Pulls nearby tickets toward the player within a limited radius. It can also appear during special events.
Ticket Booster	Temporarily multiplies only the tickets collected while the effect is active. A x2 multiplier is a reasonable starting point. It can also appear during special events.

Duplicate active power-ups should not stack. If Ticket Magnet is already active, another Ticket Magnet should not spawn until the current one is finished. Exact duration, radius, and spawn frequency still need testing.

7.4 Distance milestones

The game can acknowledge long runs with distance milestones such as 500 m, 1,000 m, 2,000 m, 5,000 m, and so on. These are not checkpoints. If the player dies, the next run still starts from zero. We can decide later whether milestones are only visual celebrations or also give small rewards/achievements.

8. Special event sections

Special events are still part of the same scrolling run. We do not want a loading screen or a separate mini-game. The player simply enters a themed part of the arcade, survives it, and then continues into the normal arcade again.

Events should be rare and last roughly 10-20 seconds. While an event is active, normal hazards such as people, towers, UFO Catcher, and Key Pusher are disabled. Tickets and approved power-ups may still appear.

8.1 Air-hockey section

The floor/environment changes into an air-hockey themed section. The main danger is the puck and possibly the pusher/striker. Normal scrolling and player movement continue.

8.2 Clown throwing-game section

The player passes through a clown/ball throwing attraction. Balls or similar projectiles become the event hazard. Like the air-hockey section, this event should have its own clear warnings and sounds.

9. Chunk generation

We do not want one fixed map because an endless game would become easy to memorize. We also do not want completely unrestricted random placement because that can create impossible or ugly layouts. The current plan is to use randomized chunk templates: each chunk has a safe structure, but several details inside it are chosen randomly each time it appears.

9.1 What a chunk is

A chunk is a reusable section of the arcade, roughly around one screen-height as a starting assumption. The next chunk is prepared above the visible screen, scrolls down through the play area, then is destroyed or recycled after it leaves the bottom.

9.2 What the chunk template controls

- Zones where static obstacles are allowed to appear.
- At least one route that must remain traversable.
- Legal spawn areas for autonomous hazards such as a person or coin tower.
- Areas where one of several ticket patterns may be placed.
- Optional power-up spawn locations.
- A hazard-budget limit and tags describing what the chunk can safely support.

9.3 What can be randomized inside the chunk

- Which allowed static object variant is used in a zone.
- Whether an optional spawn slot is filled or left empty.
- Whether the whole template is mirrored left-to-right when mirroring is safe.
- Which ticket formation is used.
- Which legal autonomous hazard is selected within the current hazard budget.
- Safe variations of hazard behavior, for example which side a person comes from, whether they stop, and whether they use their one allowed reversal.
- Cross-boundary checks so two good chunks do not accidentally make a bad combination when joined together.

This means players may eventually recognize the general shape of a chunk, but they should not know the exact solution every time. That kind of familiarity is fine; it lets experienced players improve without turning the game into a fully memorized script.

9.4 What should stay outside the chunk system

UFO Catcher and Key Pusher should be scheduled by a separate hazard-control system rather than being hard-coded into individual chunks. They target the player over time and may continue while the world is moving between chunks, so it is safer to manage them globally.

10. Hazard budget and fairness rules

10.1 Hazard budget

A hazard budget gives us a simple way to control how much danger can appear at once. Each hazard gets a rough cost. These costs are only starting values for design and testing:

Hazard	Starting cost
Moving person	1
Coin tower	2
UFO Catcher	3
Key Pusher	3

For example, a budget of 4 could allow UFO + one person, or one tower + two people if the other rules allow it. The budget alone is never enough: a combination also has to pass the compatibility and safe-route rules.

10.2 Compatibility rules

- UFO Catcher + Key Pusher: not allowed.
- UFO Catcher + person: allowed if the layout remains fair.
- UFO Catcher + coin tower: allowed if the layout remains fair.
- Key Pusher + person: allowed if the layout remains fair.
- Person + coin tower: allowed if the layout remains fair.
- Normal hazards during a special event: not allowed.

10.3 Lock-on cooldown

UFO Catcher and Key Pusher should share one cooldown. A starting range of around 10-20 seconds after a lock-on attack seems reasonable. Using a shared cooldown prevents one lock-on attack from ending and the other one starting immediately afterward.

10.4 Safe-route rule

Random generation should not create a situation where the player has no possible escape because of bad luck. There should be at least one reasonable route if the player reacts correctly. The game can still punish the player for choosing a bad position earlier, but it should not spawn a guaranteed hit with no warning.

10.5 Top-edge rule

Objects and hazards should enter from above the visible screen instead of appearing directly on top of the player. Staying near the top is still risky because the player has less reaction time, and that is an intentional trade-off. We just do not want zero-warning damage caused by a spawn happening inside the player hitbox.

11. UI, warnings and audio

11.1 Gameplay HUD

- Score / distance.
- Tickets collected in the current run (starts at 0 each run).
- Protection state, mainly shown by the character appearance; a small extra icon can be added later if needed.
- Power-up icon/timer only while a temporary power-up is active.

11.2 No tutorial

The first version will not have a separate tutorial. The game should teach the player by showing hazards in readable situations and by using the same warning cues every time. Early encounters should be easier so the player can learn what a UFO lock, key target, or wobbling tower means through play.

11.3 Audio cues

Audio is especially useful because the screen is meant to become busy. Each major hazard should have a sound that the player can learn even when they are looking somewhere else on the screen.

System	Sound idea
UFO Catcher	Mechanical whirr, targeting beeps, a clear lock sound, then claw movement.
Key Pusher	Target/key sound, clicking or ticking, lock cue, heavy smash.
Coin tower	Light coin rattle that grows stronger before the fall, then a crash.
Moving person	Footsteps and/or a small voice cue.
Ticket / power-up	Bright pickup sound that is clearly different from danger sounds.
Scalper zone	Crowd/grab sounds. The sound can become more noticeable as the player gets close to the bottom.

12. Pause and game-over flow

12.1 Manual pause

Manual pause should stop everything: world scrolling, hazard timers, power-up timers, animation, and score/distance gain.

12.2 App switching or phone lock

If the game loses focus because the player switches apps or locks the phone, the run should pause automatically. When the player returns, a short 3-2-1-GO countdown should happen before the game resumes.

12.3 Notifications

A notification banner by itself should not pause the run if the game is still active. If the player taps the notification and leaves the game, the normal auto-pause behavior applies.

12.4 Game Over

13. The final hit happens while the item is already bare.
14. A short capture animation plays, with the scalpers finally grabbing/carrying away the item.
15. A result screen shows Score, Distance, Tickets Collected, and the best record.
16. Tickets from the run are added to the saved wallet.
17. A large Retry button starts a new run quickly.

There is no revive/continue system in the first version. We can add one later if the game needs it.

13. First playable version (MVP)

The first prototype should prove that the main loop is fun before we build every hazard and cosmetic system. The first playable version should include:

- Portrait play area with mobile joystick and WASD support.
- Free 2D movement and screen boundary.
- Continuous downward world scrolling with a basic speed increase and cap.
- At least one static blocker that can push the player downward.
- Bag -> tag -> bare item -> capture protection system with 1.5-second I-frames.
- Bottom scalper zone.
- One moving person hazard.
- UFO Catcher as the first lock-on hazard.
- Tickets, a run ticket counter, and saved total tickets.
- Ticket Magnet.
- Distance-based score.
- A small set of randomized chunk templates with a safe-route check.
- Basic capture/result screen and fast Retry.

Key Pusher, coin tower, full event sections, the cosmetic shop, more power-ups, polished milestone effects, and advanced difficulty/DDA can wait until the basic run already feels good.

14. Decisions that are fixed for now

- Endless run; no normal win condition.
- Portrait phone layout.
- Free 2D movement; no lane system.
- Mobile joystick and PC WASD.
- World scrolls downward; player can move upward faster than the scroll.
- Static obstacles block/push but do not deal damage.
- Damaging hazards remove one protection layer.
- Protection order: bag -> tag -> bare item -> captured.
- 1.5 seconds of I-frames after damage.
- Bottom scalpers remove one protection layer unless the item is already bare.
- UFO Catcher and Key Pusher cannot overlap.
- One moving person at a time in the first version; around 2-second stop; at most one reversal.
- Rare event sections last around 10-20 seconds and disable normal hazards.
- Score is based on progress/distance; tickets do not add competitive score.
- Tickets are persistent currency but the visible in-run counter starts from 0.
- Current power-ups: Ticket Magnet and Ticket Booster; duplicate active effects do not stack.
- Distance milestones are not checkpoints.
- No separate tutorial.
- No revive in the first version.

15. Things we still need to decide

- Exact scroll acceleration and maximum speed.
- Exact player speed, acceleration/smoothing, joystick sensitivity, and hitbox size.
- Exact UFO/Key Pusher warning timing and shared cooldown after playtesting.
- Exact Ticket Magnet radius/duration and Ticket Booster duration.
- Power-up spawn rules and whether we need a pity/guarantee system after a long gap.
- Exact chunk dimensions and how chunk connections are represented in code.
- How many starter chunk templates we need. A first target of about 6-10 is reasonable.
- The exact algorithm used to check safe routes across two neighboring chunks.
- Whether milestone messages give only celebration feedback or also rewards.
- Cosmetic shop items and prices.
- Final art perspective: pure top-down art or top-down gameplay with a mild faux-perspective look.
- What happens to difficulty after scroll speed reaches the cap. This can be designed together with the later DDA/research work.

16. Terms we will use

Term	Meaning in this project
World scrolling	The arcade environment moves downward to create the feeling of forward movement.
Control area	The part of the portrait screen where the player is allowed to move.
Static obstacle / blocker	Solid non-damaging object that can stop or push the player.
Damaging hazard	Anything that removes one protection layer on contact.
Autonomous hazard	Hazard that follows its own movement/state pattern instead of continuously targeting the player.
Lock-on hazard	Hazard that tracks the player, locks a target, then attacks that fixed target.
Telegraph	Warning before an attack.
Lock / commit	Moment when a tracking attack stops following the player and fixes its final target.
I-frames	Temporary invincibility after a hit; currently 1.5 seconds.
Protection state	The current bag/tag/bare-item life state.
Chunk	Reusable section/template of the endless arcade.
Constrained procedural generation	Randomizing content inside rules/templates instead of placing everything freely at random.
Hazard budget	A simple number used to limit how much danger can be active together.
Compatibility rule	Rule that allows or forbids a hazard combination even if the budget would allow it.
Safe-route rule	Rule that prevents random generation from creating an unavoidable no-win layout.
Event section	Rare 10-20 second themed part of the scrolling run with its own hazard.
Run tickets	Tickets collected in the current attempt. This counter starts at 0 each run.
Saved tickets	Persistent ticket wallet used for later in-game purchases.
Distance milestone	Progress marker such as 500 m or 1,000 m; not a checkpoint.

17. Reference games

These games are references for individual ideas and genre conventions. We are not trying to copy their full gameplay. Our main differences are free 2D movement, the bag/tag protection idea, the arcade-building theme, randomized chunk templates, and player-targeting lock-on hazards.

[1] Subway Surfers - Apple App Store / Apple editorial

Reference for the endless chase feeling, obstacle dodging, collectible currency, responsive mobile controls, high-score play, and crash-protection ideas such as hoverboards.

https://apps.apple.com/us/iphone/story/id1453245869

[2] Subway Surfers Help Center - Power-Ups

Reference for in-run pickups such as Coin Magnet. Our Ticket Magnet uses the same broad idea, while our Ticket Booster affects currency rather than competitive score.

https://sybo.helpshift.com/hc/en/5-subway-surfers/faq/209-power-ups/

[3] Subway Surfers Help Center - Coins

Reference for collecting currency during a run and keeping that currency for later use.

https://sybo.helpshift.com/hc/en/5-subway-surfers/faq/147-coins/

[4] Subway Surfers Help Center - Hoverboards & their Powers

Reference for the idea of surviving a crash through a protection system. Our version uses three packaging states instead of a temporary board.

https://sybo.helpshift.com/hc/en/5-subway-surfers/faq/208-hoverboards-their-powers/

[5] Temple Run - Imangi Studios

Reference for endless escape, reflex-based obstacle avoidance, collectible currency, power-ups, unlocks, and distance/high-score chasing.

https://imangistudios.com/thegames/temple-run/

[6] Temple Run FAQ - Scoring

Reference for how an endless runner can combine movement/progress and other actions in a score. We are deliberately keeping our competitive score simpler and based on progress so ticket RNG does not affect the leaderboard.

https://imangistudios.com/thefaq/how-does-scoring-work-in-temple-run/

[7] Jetpack Joyride - Halfbrick

Reference for persistent coin collection and Coin Magnet-style effects in an endless arcade-action game.

https://www.halfbrick.com/blog/in-depth-jetpack-joyride-gadgets

[8] Crossy Road - Hipster Whale Support

Reference for persistent unlock/progression ideas and mobile interruption/pause behavior.

https://www.hipsterwhale.com/crossy-road-support

[9] Flappy Bird - general historical reference

Reference for a very simple progress-based arcade score and very fast restart loop. We use it only as a basic genre comparison; our game is more forgiving because it has three protection states.

https://en.wikipedia.org/wiki/Flappy_Bird

Next design session

The next useful step is to design the chunk system in more detail: chunk size, spawn zones/sockets, how we represent the safe corridor, how two neighboring chunks are checked together, and around 6-10 starter chunk templates. Once that is clear, we can start turning the document into implementation tasks.